UNIT 1 · TOPIC 1.3

Best Practices for Public Networks

Assignment, Answer Key

Teacher answer key, 39 points total.

Practice Questions

1.3.a · Adversary skill levels (low-skilled vs high-skilled / zero-days); motivation

1. An attacker at a coffee shop runs a freely downloaded packet-capture script without understanding how it works. Which tier best describes this adversary? (1 pt)

- ○ A. High-skilled
- ○ B. Nation-state
- **● C. Low-skilled**
- ○ D. Zero-day operator

Why: Using off-the-shelf tools without understanding the mechanics is the defining characteristic of a low-skilled adversary (script kiddie).

2. A zero-day vulnerability is dangerous primarily because: (1 pt)

- ○ A. It only works on Windows systems
- ○ B. It requires a physical connection to exploit
- ○ C. It can only be used by government agencies
- **● D. The vendor has not yet released a patch, so defenders have no fix available**

Why: Zero-day means zero days of vendor warning, no patch exists, leaving all users of the affected software exposed.

3. Which of the following motivations would lead an attacker to target a SPECIFIC high-value individual rather than sweeping widely? (Select ALL that apply.) (2 pts)

- □ A. Financial gain (bulk credential resale)
- **■ B. Espionage (stealing trade secrets)**
- □ C. Curiosity / learning
- **■ D. Targeted corporate sabotage**

Why: Espionage and targeted sabotage require a specific victim. Bulk credential resale and curiosity are typically opportunistic and sweep broadly.

4. Name ONE observable behavior that would place an attacker firmly in the high-skilled tier. (1 pt)

Acceptable answers: zero-day; custom tool; writes own tool; modifies tool; adapts to defenses; exploit development

Why: High-skill markers include zero-day exploitation, custom tool development, and the ability to adapt when initial attacks fail.

5. Why does the skill tier of an adversary matter for a risk assessment? (1 pt)

- ○ A. Higher skill means the attacker is always caught faster
- **● B. Skill tier determines the breadth and precision of possible attacks, shaping both likelihood and severity**
- ○ C. Low-skilled attackers always cause more damage
- ○ D. Skill tier has no effect on the attacks they can perform

Why: Low-skilled attacks are broad but shallow; high-skilled attacks are precise and deep. Skill tier directly changes both how many victims are reachable and how severe the harm can be.

1.3.b · Evil twin attack

6. What does BSSID stand for and why is it relevant to detecting an evil twin? (1 pt)

○ A. Broadband Service Signal ID, it tells you the ISP providing the connection

○ B. Basic Security Standard ID, it encodes the encryption type in use

○ C. Broadcast SSID, it is the human-readable name the user sees in the network list

● D. Basic Service Set Identifier, it is the hardware MAC address of the AP radio, unique per physical device

Why: BSSID is the MAC address of the physical radio broadcasting the network. Two APs cannot share the same BSSID legitimately; an evil twin will have a different BSSID from the real AP even if the SSID matches.

7. A wireless scan shows two entries: both broadcast SSID 'AirportFree', one with security WPA2 and one with OPEN. Select ALL indicators that suggest an evil twin is present. (2 pts)

■ A. The same SSID is advertised by more than one BSSID

■ B. One entry has weaker (or no) security than the other for the same SSID

□ C. Both entries have the same signal strength

□ D. The network name contains the word 'Free'

Why: Duplicate SSID across multiple BSSIDs is a primary evil-twin indicator. A security downgrade (WPA2 vs. OPEN) for the same SSID is a secondary indicator. Signal strength and network name alone are not definitive.

8. Which attack does an evil twin primarily enable once the victim connects? (1 pt)

○ A. SQL injection against the victim's database

● B. Man-in-the-middle, the attacker relays and can read or alter all victim traffic

○ C. Distributed denial-of-service against a remote server

○ D. Brute-force of the victim's Wi-Fi password

Why: Once connected to the evil twin, all victim traffic flows through the attacker's hardware, enabling a man-in-the-middle position.

9. An enterprise legitimately deploys three access points all broadcasting SSID 'CorpNet' with three different BSSIDs. A fourth entry appears with SSID 'CorpNet' and a BSSID not on the approved list. What does this indicate? (1 pt)

○ A. Normal, any AP can join a corporate SSID

○ B. The network is using WPA3, which always shows extra BSSIDs

○ C. A misconfigured legitimate AP that should be rebooted

● D. A potential evil twin; the unknown BSSID is not an approved radio

Why: An unapproved BSSID broadcasting an authorized SSID is the classic WIDS alert for an evil twin. Multiple BSSIDs per SSID are normal in enterprise deployments, but each BSSID must be on the approved list.

10. Name two defenses an END USER (not an administrator) can use to reduce evil-twin risk. (2 pts)

Acceptable answers: vpn; https; avoid auto-connect; verify ssid; ask staff; cellular data; check certificate

Why: End-user defenses include: using a VPN to encrypt all traffic regardless of the network, always checking for HTTPS, disabling auto-connect to open networks, and confirming the correct SSID with venue staff before connecting.

1.3.c · Jamming / denial-of-service attack

11. Which Wi-Fi access mechanism does RF jamming exploit to deny service? (1 pt)

- ○ A. WPA3 handshake, jamming corrupts the 4-way handshake
- **● B. CSMA/CA, devices back off when they detect channel noise, including deliberate jamming**
- ○ C. DHCP, jamming prevents IP address assignment
- ○ D. DNS, jamming reroutes domain lookups

Why: Wi-Fi uses CSMA/CA: devices listen before transmitting and back off when they detect noise. A jammer floods the channel with noise, causing all devices to wait indefinitely.

12. What distinguishes RF jamming from ordinary transient interference (e.g. a nearby microwave)? (1 pt)

- ○ A. Jamming always operates on 5 GHz; microwaves only affect 2.4 GHz
- ○ B. Only jamming uses radio waves; interference uses wired signals
- ○ C. Jamming raises RSSI; interference lowers it
- **● D. Jamming produces sustained, high-level noise across many consecutive samples; interference is typically brief**

Why: The detection key is duration: a jammer maintains elevated noise over many consecutive readings, while household interference (microwave, Bluetooth) causes momentary spikes.

13. Select ALL defenses that can help a network survive or detect a jamming attack. (3 pts)

- **■ A. Switch to a different Wi-Fi channel not targeted by the jammer**
- □ B. Enable WPA3 encryption
- **■ C. Deploy a WIDS configured to alert on sustained noise-floor anomalies**
- **■ D. Move from 2.4 GHz to 5 GHz or 6 GHz to use channels with shorter jammer range**

Why: Channel switching and frequency-band migration reduce exposure to a narrow-band jammer. A WIDS can detect sustained noise anomalies. WPA3 encrypts data but provides no protection against RF jamming, which operates at the physical layer below encryption.

14. In the coding problem, solve(samples) returns True when noise samples indicate jamming. Which rule does it apply? (1 pt)

- ○ A. Any single sample exceeds the threshold
- ○ B. The average of all samples exceeds the threshold
- ○ C. More than half the samples exceed the threshold
- **● D. At least N consecutive samples all exceed threshold T**

Why: The consecutive-sample rule (N or more samples in a row above T) detects sustained interference while ignoring brief transient spikes.

15. Name the metric, always a negative number, that wireless monitors use to measure signal power. (1 pt)

Acceptable answers: rssi; received signal strength; dbm; signal strength indicator

Why: RSSI (Received Signal Strength Indicator), measured in dBm (always negative for Wi-Fi; e.g. -55 dBm).

1.3.d · War driving; protecting yourself (verify SSIDs, avoid open networks, VPN)

16. What does a war driver primarily collect about nearby Wi-Fi networks? (1 pt)

- ○ A. User passwords transmitted over the networks
- **● B. SSIDs, BSSIDs, security types, signal strengths, and GPS locations**
- ○ C. The names of devices connected to each network
- ○ D. The browsing history of users on each network

Why: War driving collects network-level metadata: SSID (name), BSSID (hardware address), security protocol, signal strength, and location. Passwords and browsing history require active attacks beyond passive scanning.

17. Select ALL factors that a personal risk-assessment protocol should check before connecting to a public Wi-Fi network. (3 pts)

- ■ **A. Whether the SSID has been verified with venue staff**
- ■ **B. Whether the network is open (no password)**
- □ C. The brand of the router hardware
- ■ **D. Whether you have an active VPN**

Why: The three-step protocol: (1) verify the SSID, (2) avoid open networks, (3) use a VPN. Router brand is not a user-accessible or reliable risk indicator.

18. Why does SSID verification come FIRST in the personal Wi-Fi safety protocol, before VPN activation? (1 pt)

- ○ A. VPNs are slower than SSID checks
- ○ B. Venue staff can configure the VPN for you
- ● **C. Connecting to the wrong network (evil twin) sends traffic to an adversary before the VPN tunnel is established**
- ○ D. Open networks cannot run VPN software

Why: If you connect to a rogue network first, even a brief unencrypted window before the VPN activates can leak credentials. Verification prevents the initial connection from being adversarial.

19. An attacker uses a public war-driving database to learn a corporation's SSID, then includes it in a phishing email. Which attack combination does this illustrate? (1 pt)

- ○ A. SQL injection combined with brute-force
- ● **B. Physical-layer reconnaissance combined with social engineering**
- ○ C. Jamming combined with man-in-the-middle
- ○ D. Port scanning combined with evil twin

Why: War driving provides physical-layer intelligence (SSID) that increases the plausibility of a social-engineering attack, illustrating how passive reconnaissance feeds active attacks.

20. Name the one protection that defends you against interception even if the public Wi-Fi network is legitimate but compromised. (1 pt)

Acceptable answers: vpn; virtual private network; encrypted tunnel; tunnel

Why: A VPN encrypts traffic before it leaves your device, so even a compromised or malicious access point sees only ciphertext.

Coding Assignments (Python 3)

1.3.a · Adversary skill levels (low-skilled vs high-skilled / zero-days); motivation

3 pts · auto-graded on the site

Coding problem, 1.3.a: Classify adversary skill tier

Skill: Analyze Risk · **Maps to:** 1.3.A · **Language:** Python 3 (pytest)

Problem

Security analysts use observable attacker behaviors to infer capability level. Given a profile of an adversary's tools and behaviors, classify them as either a **low-skilled** or **high-skilled** threat actor.

Function contract

Implement `solve(adversary: dict) -> str` in `starter.py` (see `interface.md` / `blackbox.yaml`). Return `"low"` or `"high"` based on the classification rules defined in `interface.md`.

Input

A dict with exactly these boolean keys:

- `uses_zero_day`, exploits an unpublished vulnerability
- `custom_tooling`, writes or substantially modifies their own tools
- `targets_specific_victim`, selects a specific high-value target
- `off_the_shelf_only`, relies solely on public tools without modification
- `adapts_to_defenses`, changes tactics when initial attempts fail

Output

The string `"low"` or `"high"`.

Classification rules (from interface.md)

1. If `uses_zero_day` OR `custom_tooling` -> `"high"` 2. If `targets_specific_victim` AND `adapts_to_defenses` -> `"high"` 3. Otherwise -> `"low"`

Examples

```
{"uses_zero_day": True,  "custom_tooling": False, "targets_specific_victim": False,
 "off_the_shelf_only": False, "adapts_to_defenses": False}  ->  "high"

{"uses_zero_day": False, "custom_tooling": False, "targets_specific_victim": False,
 "off_the_shelf_only": True,  "adapts_to_defenses": False}  ->  "low"

{"uses_zero_day": False, "custom_tooling": False, "targets_specific_victim": True,
 "off_the_shelf_only": False, "adapts_to_defenses": True}   ->  "high"
```

Constraints

- Deterministic, offline, standard library only.
- Must pass all cases in `tests.py` (visible + hidden).

How to run

```
cd problem && python -m pytest tests.py -q
```

Reference solution:

```
"""Reference solution for 1.3.a. MUST pass 100% of tests.py.
Also the default 'blackbox' implementation students can swap out (see blackbox.yaml)."""


def solve(adversary: dict) -> str:
    """Classify an adversary as 'low' or 'high' skill based on their profile.

    Rules (first match wins):
      1. uses_zero_day OR custom_tooling                        -> 'high'
      2. targets_specific_victim AND adapts_to_defenses         -> 'high'
      3. otherwise                                              -> 'low'
```

```
    """
    if adversary["uses_zero_day"] or adversary["custom_tooling"]:
        return "high"
    if adversary["targets_specific_victim"] and adversary["adapts_to_defenses"]:
        return "high"
    return "low"
```

1.3.b · Evil twin attack

3 pts · auto-graded on the site

Coding problem, 1.3.b: Detect evil-twin access points

Skill: Analyze Risk, Detect Attacks · **Maps to:** 1.3.B · **Language:** Python 3 (pytest)

Problem

A wireless scanner returns a list of detected access points. Each AP is a dict with keys:

- `ssid` (str), the human-readable network name
- `bssid` (str), the hardware MAC address of the radio (normalized to uppercase)
- `security` (str), one of `"OPEN"`, `"WEP"`, `"WPA"`, `"WPA2"`, `"WPA3"`
- `signal` (int), received signal strength in dBm (negative; stronger = closer to 0)

An AP is **suspicious** (potential evil twin) if ANY of the following holds:

1. **Duplicate SSID / multiple BSSIDs**, the same SSID is being broadcast by more than one BSSID in the scan. Every BSSID in such a group is flagged. 2. **Security downgrade**, an SSID appears with at least two different security levels and the AP in question does not have the strongest security level seen for that SSID. Those weaker entries are flagged.

Rules combine: an AP is flagged if either condition is true for it.

Function contract

Implement `solve(aps: list[dict]) -> list[str]` in `starter.py` (see `interface.md` / `blackbox.yaml`).

Return a **sorted, de-duplicated** list of suspicious BSSIDs.

Input

A list of dicts, each with keys `ssid`, `bssid`, `security`, `signal`. The list may be empty. BSSIDs are already uppercase.

Output

A sorted list of BSSIDs (strings) that are suspicious. Empty list if none.

Security ordering (weakest to strongest)

OPEN < WEP < WPA < WPA2 < WPA3

Examples

```
# Two APs share SSID "CafeGuest" — different BSSIDs -> both flagged (rule 1)
# The OPEN entry also has weaker security (rule 2)
aps = [
    {"ssid": "CafeGuest", "bssid": "AA:BB:CC:11:22:33", "security": "WPA2", "signal": -55},
    {"ssid": "CafeGuest", "bssid": "DE:AD:BE:EF:00:01", "security": "OPEN", "signal": -42},
    {"ssid": "HotelLobby","bssid": "11:22:33:44:55:66", "security": "WPA2", "signal": -65},
]
solve(aps) -> ["AA:BB:CC:11:22:33", "DE:AD:BE:EF:00:01"]
```

```
# Empty scan
solve([]) -> []

# Single SSID, single BSSID, no anomaly
solve([{"ssid": "HomeNet", "bssid": "00:11:22:33:44:55", "security": "WPA3", "signal": -50}]) -> []
```

Constraints

- Deterministic, offline, standard library only.
- Must pass all cases in `tests.py` (visible + hidden).

How to run

```
cd problem && python -m pytest tests.py -q
```

Reference solution:

```
"""Reference solution for 1.3.b. MUST pass 100% of tests.py.
Also the default 'blackbox' implementation students can swap out (see blackbox.yaml)."""
from typing import List

# Security strength: higher index = stronger
_SEC_ORDER = ["OPEN", "WEP", "WPA", "WPA2", "WPA3"]


def _sec_rank(sec: str) -> int:
    try:
        return _SEC_ORDER.index(sec.upper())
    except ValueError:
        return -1


def solve(aps: List[dict]) -> List[str]:
    """Return a sorted, de-duplicated list of suspicious BSSIDs.

    Rule 1 — Duplicate SSID: SSID seen with >1 distinct BSSID -> flag all BSSIDs for it.
    Rule 2 — Security downgrade: SSID seen with multiple security levels ->
              flag any BSSID whose security is not the maximum for that SSID.
    """
    if not aps:
        return []

    # Group by SSID
    from collections import defaultdict
    ssid_entries = defaultdict(list)  # ssid -> list of (bssid, security)
    for ap in aps:
        ssid_entries[ap["ssid"]].append((ap["bssid"], ap["security"]))

    suspicious = set()

    for ssid, entries in ssid_entries.items():
        bssids = [e[0] for e in entries]
        secs = [e[1] for e in entries]

        # Rule 1: more than one BSSID for this SSID
        if len(set(bssids)) > 1:
            suspicious.update(bssids)

        # Rule 2: security downgrade — flag entries without max security
        ranks = [_sec_rank(s) for s in secs]
        max_rank = max(ranks)
        if len(set(ranks)) > 1:
```

```
            for bssid, rank in zip(bssids, ranks):
                if rank < max_rank:
                    suspicious.add(bssid)

    return sorted(suspicious)
```

1.3.c · Jamming / denial-of-service attack

3 pts · auto-graded on the site

Coding problem, 1.3.c: Detect sustained wireless jamming

Skill: Analyze Risk · **Maps to:** 1.3.B · **Language:** Python 3 (pytest)

Problem

A wireless intrusion detection system samples the noise floor of a Wi-Fi channel once per second. Each sample is an integer noise measurement (higher = more noise). Your job is to detect whether the sample series contains a sustained jamming event.

Detection rule: a series indicates jamming if it contains a run of at least **N = 5** consecutive samples that are ALL strictly greater than **T = 80**.

These constants are fixed for this problem (see `interface.md`). A single spike, or a short burst of noise below the window, is not considered jamming.

Function contract

Implement `solve(samples: list[int]) -> bool` in `starter.py` (see `interface.md` / `blackbox.yaml`).

Return `True` if the series contains a sustained jamming event; `False` otherwise.

Input

A list of integers (noise-floor samples). May be empty. No length limit. Values >= 0.

Output

`True` (jamming detected) or `False` (no sustained jamming).

Examples

```
[20, 25, 90, 22, 30, 18, 24]            -> False  (only one spike; run length 1 < N=5)
[15, 82, 85, 88, 91, 83, 87, 20]        -> True   (positions 1-6 all > 80; run = 6 >= N=5)
[70, 72, 75, 71, 73, 74, 76]            -> False   (none exceed T=80)
[50, 81, 82, 83, 84, 85, 86, 87, 88, 30]-> True   (positions 1-8 all > 80; run = 8 >= N=5)
[]                                       -> False  (empty input)
```

Constraints

- T = 80 (strictly greater than), N = 5 (minimum consecutive samples above T).
- Deterministic, offline, standard library only.
- Must pass all cases in `tests.py` (visible + hidden).

How to run

```
cd problem && python -m pytest tests.py -q
```

Reference solution:

```
"""Reference solution for 1.3.c. MUST pass 100% of tests.py.
Also the default 'blackbox' implementation students can swap out (see blackbox.yaml)."""
```

```
from typing import List

THRESHOLD = 80   # noise floor threshold (strictly greater than)
WINDOW = 5       # minimum consecutive samples above threshold to flag jamming


def solve(samples: List[int]) -> bool:
    """Return True if the sample series contains a sustained jamming event.

    A series contains sustained jamming if it has a run of at least WINDOW
    consecutive samples each strictly greater than THRESHOLD.
    """
    run = 0
    for s in samples:
        if s > THRESHOLD:
            run += 1
            if run >= WINDOW:
                return True
        else:
            run = 0
    return False
```

1.3.d · War driving; protecting yourself (verify SSIDs, avoid open networks, VPN)

3 pts · auto-graded on the site

Coding problem, 1.3.d: Public Wi-Fi risk flagger

Skill: Mitigate Risk · **Maps to:** 1.3.B, 1.3.C · **Language:** Python 3 (pytest)

Problem

You are building the risk-assessment routine a network-policy tool runs before letting a device connect to a public Wi-Fi network. Given a description of one network and the user's current posture, return the **risk flags** that apply, so the user (or an administrator) can make an informed decision. This is exactly the three-step personal protocol from the lesson, expressed as code: verify the SSID, avoid open networks, always use a VPN.

The function is pure and deterministic, no network, no I/O.

Function contract

Implement `solve(network: dict) -> list[str]` in `starter.py` (see `interface.md` / `blackbox.yaml`).

Input

`network`, a `dict` describing one network and the user's posture toward it. Recognized keys:

Key	Type	Meaning
`"encryption"`	`str`	Wi-Fi security type, e.g. `"open"`, `"wpa2"`, `"wpa3"`. Case-insensitive. Missing / None / `""` is treated as open (worst case).
`"verified"`	`bool`	`True` if the SSID was confirmed with venue staff / an official source. Missing is treated as `False`.
`"vpn"`	`bool`	`True` if a VPN tunnel is active on the device. Missing is treated as `False`.

Example:

```
{"ssid": "AirportFreeWifi", "encryption": "open", "verified": False, "vpn": False}
```

Output

A **sorted, de-duplicated** `list[str]` of the risk flags that apply, drawn from:

- `"open-network"`, the network is open (no Wi-Fi-layer encryption).
- `"unverified-ssid"`, the SSID has not been confirmed with an authoritative source.

- `"no-vpn"`, no VPN is protecting the user's traffic.

Return `[]` when none apply (a verified, encrypted network used over a VPN).

Examples

```
{"encryption": "open", "verified": False, "vpn": False}   -> ["no-vpn", "open-network", "unverified-ssid"]
{"encryption": "wpa2", "verified": True,  "vpn": True}     -> []
{"encryption": "wpa2", "verified": True,  "vpn": False}    -> ["no-vpn"]
{"encryption": "WPA3", "verified": False, "vpn": True}     -> ["unverified-ssid"]
{}                                                          -> ["no-vpn", "open-network", "unverified-ssid"]
```

Constraints

- Deterministic, offline, standard library only.
- Matching of `encryption` is case-insensitive.
- Output always sorted ascending and de-duplicated.
- Must pass all cases in `tests.py` (visible + hidden).

How to run

```
cd problem && python -m pytest tests.py -q
```

Reference solution:

```
"""Reference solution for 1.3.d. MUST pass 100% of tests.py.
Also the default 'blackbox' implementation students can swap out (see blackbox.yaml).

Applies the three-step public-Wi-Fi protocol as risk flags: verify the SSID, avoid open
networks, always use a VPN. Pure, deterministic, stdlib only."""
from typing import List

OPEN_VALUES = {"open", "none", ""}


def solve(network: dict) -> List[str]:
    """Return the sorted, de-duplicated risk flags for one network description.

    Flags: "open-network", "unverified-ssid", "no-vpn".
    See prompt.md / interface.md for the rules.
    """
    network = network or {}
    flags = set()

    encryption = network.get("encryption")
    if encryption is None or str(encryption).strip().lower() in OPEN_VALUES:
        flags.add("open-network")

    if not network.get("verified", False):
        flags.add("unverified-ssid")

    if not network.get("vpn", False):
        flags.add("no-vpn")

    return sorted(flags)
```